

Master Thesis

Tracing the coronary arteries in 3D computed tomography scans

Weekly Report 2

Malgorzata Korona. s223322
Mateusz Baldachowski. s230241

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1 This week's progress

1.1 Introduction

Cardiovascular diseases (CVDs) are the most common global cause of death. Data gathered by WHO [1] indicate that in 2019 alone almost 18 million people died due to CVD which accounted for around 1/3 of global deaths. Among those 85% were caused by either heart attack or stroke. Both of them are usually caused by disruption in blood flowing to the heart or brain.

Disease of blood vessels responsible for supplying the heart muscle, called a coronary heart disease, is directly linked to coronary artery which is the major blood vessel pumping the blood into the heart. Patients suffering from it have a disrupted blood flow caused by accumulated build-ups of fatty deposits on the inner walls of the coronary artery. Diagnostics of coronary artery disease (CAD) is therefore crucial to guide appropriate treatment in order to prevent heart failures.

One way to diagnose CAD is through a coronary computed tomography angiography (CCTA). This modality has been proven to be effective and since it is a non-invasive procedure it has been widely recommended. The problem with it, however, is that mapping of the coronary artery based on the 3D image volume is usually done manually by the experts which makes it time-consuming and error-prone. Building on the previous contributions of a peer in our research group ??, this study aims to enhance the algorithm used for automating coronary artery tracing. Furthermore, the project explores the classification of the coronary artery into sections aligned with the conventions used by cardiovascular surgeons and specialists in the field ?. Finally, the research delves into estimation of the coronary tree topology "phenotype" to see if there are shared characteristics of a coronary artery anatomy among patients ??.

1.2 Heart anatomy

Heart is an organ that is responsible for supplying blood to the body. It consists of four chambers: **right atrium (RA)**, **left atrium (LA)**, **right ventricle (RV)** and **left ventricle (LV)**. The atria are smaller than the ventricles. RA sends the deoxygenated blood into the RV after receiving it from **vena cava** - human body's largest vein. RV then pumps the blood into the pulmonary circuit, where it's oxygenated. Afterwards, oxygenated blood comes back into the LA and is then sent into LV. LV then pumps the blood into the rest of the body. At the end, the blood returns through vena cava to the right atrium and the whole cycle repeats. [2, 3] The circulatory system of a human is visible in figure 1.

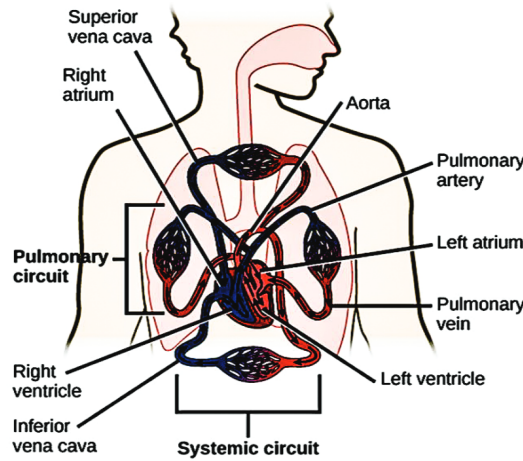


Figure 1: Circulatory system of a human, comprising systemic, pulmonary, and coronary circuits [4]. Blood enters right atrium from systemic circuit through vena cava, is sent to right ventricle and then to pulmonary circuit. Oxygenated blood comes back to cardiac circuit: first to left atrium, then to left ventricle. Finally it exits the cardiac circuit through the aorta and is sent to other organs.

The cardiac cycle in which heart receives blood and then ejects it through the arteries is divided into diastolic phase (diastole), in which heart chambers are relaxed receive blood, and systolic phase (systole), when the blood is sent out due to chambers' contractions. The atria and ventricles take turns contracting and relaxing. When the atria are in diastolic phase, the ventricles are systolic phase, and vice versa [5]. There are four valves in the heart that ensure synchronization between those phases and the correct direction of the passage of blood by appropriate opening and closing. There are two types of valves: atrioventricular valves, including the tricuspid and mitral valves, and semilunar valves, including the pulmonary and aortic valves. Atrioventricular valves connect atria with ventricles: tricuspid one on the right side of the heart, and mitral one on the left side. Semilunar valves are responsible for blood flow between ventricles and arteries: pulmonary valve is connected to the right ventricle, and aortic valve

to the left ventricle [2, 3, 6]. Passage of the blood throughout heart's chambers and valves is demonstrated in figure 4

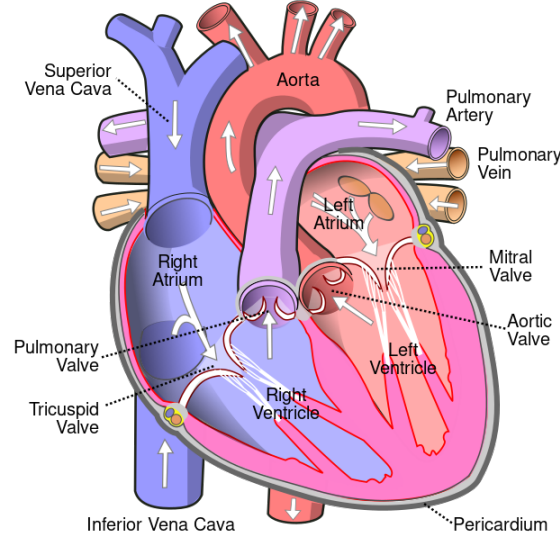


Figure 2: The passage of blood in the human's heart. [7]

Cardiac muscle, called myocardium, is responsible for contraction and relaxation of the heart, enabling it to receive and pump blood. Together with two other muscle tissues it forms a the heart wall, with myocardium being the thickest layer. Coronary arteries supply blood to the cardiac muscle, facilitating its contractile function [2].

Coronary arteries encircle the heart, with small branches extending into the myocardium to supply it with blood. There are two main coronary arteries: **left coronary artery (LCA)** and **right coronary arter (RCA)**. They branch from the base of the aorta (human body's largest artery). Specifically, right and left coronary artery arise from right and left cardiac sinuses, respectively, that are two out of three dilations in aorta's wall just at the base of the aorta [2, 3]. Coronary arteries are shown in the figure 3

Left coronary artery is a short vessel, that branches into **left anterior descending artery (LAD)** and **left circumflex artery (LCX)**. In some cases the LCA might not be present with LAD and LCX emerging directly from the left cardiac sinus. LAD, together with its branches, supplies both of the ventricles and interventricular septum (a wall separating lower chambers of the heart and ventricles). LCX mainly supplies the left atrium and parts of left ventricle. It also branches into one or more obtuse marginal (OM) arteries that supply the left ventricle as well [2, 3, 8].

The RCA branches into three major branches: the **posterior descending artery (PDA)**, supplying interventricular septum, the **nodal artery**, supplying right atrium) as well as the **right marginal artery**, that provides blood to the right ventricle and PDA [2, 3, 8].

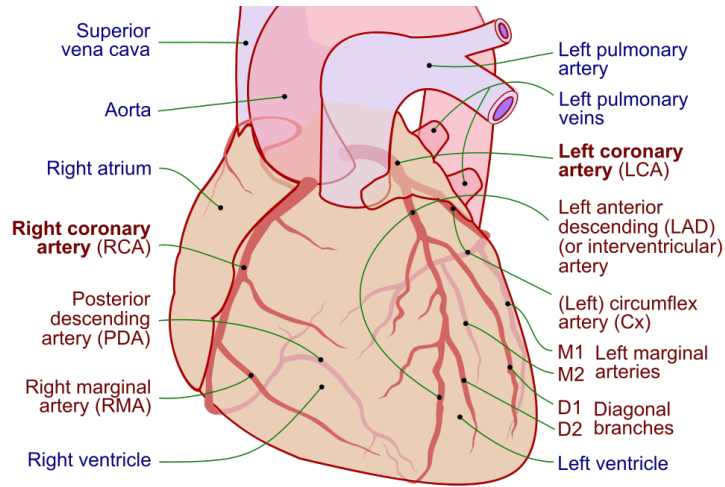


Figure 3: Coronary arteries (red text) and other key landmarks (blue text). [9]

Depending on where the PDA branches of, we define three types of heart dominance [3, 8]:

- right-heart dominance, when PDA originates from RCA [3, 8], observed in around 70% of the population [10],
- left-heart dominance, with PDA branching of LCX [3, 8], observed in 5-10% of the population [10],
- codominance, when PDA is a result of anastomosis created between LCX and RCA [3, 8], observed in the rest of the population [10].

1.3 Coronary artery disease (CAD)

Cardiovascular diseases (CVD) are the primary cause of death worldwide. According to the World Health Organization, in 2018, cardiovascular diseases, including heart disease, brain-related vascular diseases, and blood vessel disorders, were responsible for 18 million deaths [1]. CVDs can be classified in the following way [11]:

1. Caused by atherosclerosis:
 - Ischemic or coronary artery disease (e.g., heart attack)
 - Cerebrovascular disease (related to the brain – e.g., stroke)
 - Diseases of the aorta and arteries (e.g., hypertension or peripheral vascular disease)
2. Other:
 - Congenital heart disease

- Rheumatic heart disease
- Cardiomyopathies
- Cardiac arrhythmias

The most serious across the population are by far the first two caused by atherosclerosis - CAD and cerebrovascular disease which accounted for 85% of CVD related deaths in 2018. Some studies indicate that in America more than a half of aged 45 to 84 can suffer from atherosclerotic disease and may not be aware of that [12].

Atherosclerosis is a condition in which over time inner walls of arteries, blood vessels transporting the oxygenated blood from the heart chambers to the muscles, accumulate fat, cholesterol, calcium and other substances. The build-up deposits, called plaques, can narrow down the lumen limiting the blood flow. In the most severe scenario plaques can rupture and form a blood clot which obstruct the circulation.

When atherosclerosis occurs in the coronary artery it is called a coronary artery disease. As coronary arteries are responsible for supplying the heart muscle with blood together with oxygen and nutrients, their blockage can cause serious consequences. Blocked blood flow leads to myocardium being deprived of oxygen which damages it causing, often fatal, myocardial infraction (heart attack).

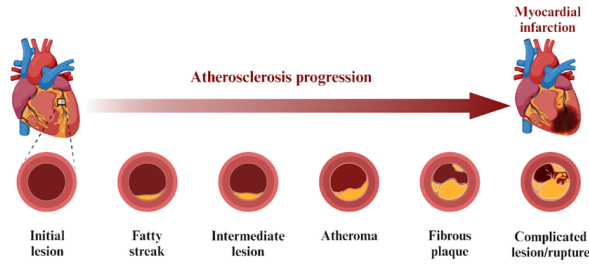


Figure 4: Atherosclerosis development [13].

Atherosclerotic disease is a slowly developing disease which often does not show any symptoms until it reaches its late stages. By that time it might be too late to prevent its serious consequences. This highlights the importance of early detection during the subclinical phase to maintain a healthy population [14].

1.4 Coronary computed tomography angiography (CCTA)

Coronary computed tomography angiography (CCTA) is one of the diagnostic modalities which has been quickly developing over the recent years. CCTA is recommended as an initial test to establish whether patient suffers from CAD according to the guidelines proposed by both European and American cardiology societies [15] [16]. Recent studies have shown that CCTA is a highly accurate non-invasive modality which has been proven to be as effective as its invasive counter-part, conventional coronary angiography (CCA) [17] [18].

2 Project status

Project is going according to the plan. Literature review needs to be finished next week, however we finished writing about heart anatomy and cardiovascular diseases.

3 Plan for the next weeks

1. Finish literature review
2. Getting familiar with Teodora's progress
3. Exploring and running modelling pipeline

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